

Noah Teo Rui-Sheng

Work authorization: Singapore Citizen | H-1B1 Visa Eligible (Singapore-specific Fast-track Visa to the US, No Lottery Required)

Mobile: +6596548995 | Email: noahteo999@gmail.com

Linkedin: www.linkedin.com/in/noah-teo | Github: <https://github.com/naozitios>

EDUCATION

National University of Singapore (NUS) School of Computing, BSc (Honors) in Business

2020 Aug - 2024 June

Analytics, double specialization in Financial Analytics & Machine Learning

CFA Level I Candidate (May 2026)

WORK EXPERIENCE

JPMorgan Chase & Co Cloud Data Engineer - (public cloud inventory)

2024 Aug - Present

- **Distributed systems engineering:** Engineered a distributed caching layer using Redis (AWS ElastiCache), enabling shared cache access across instances; reduced external API calls by 10× and improved request latency by 10×
- **Data Warehouse Architecture:** Designed and developed a scalable data warehouse for Atlas (JPMC's cloud provisioning platform), supporting 50,000+ teams and approximately 150M cloud resources. Enabled reliable inventory ingestion and reporting for C-suite stakeholders
- **ETL & Data Pipelines:** Built end-to-end ETL pipelines using AWS EventBridge, Glue, and Redshift to ingest, transform, and store large-scale infrastructure data. Owned multiple pipelines, including scheduled API crawls from the gateway to S3, schema management via Liquibase, and data transformations into Redshift
- **Data Access & Visualization:** Developed APIs and frontend visualizations for data exploration and reporting using React and Plotly, enabling internal teams to easily analyze and consume platform metrics
- **Observability & Reliability:** Implemented observability and alarm systems on AWS using Terraform and Datadog, enabling proactive alerting for potential system failures and reducing incident resolution time by 50%
- **Secrets Management & Scalability:** Implemented shared secret management and automated rotation using AWS Secrets Manager, enabling multiple services to securely consume the same credentials while supporting horizontal scaling. Increased maximum system load capacity by at least 10×, with further scalability achievable through auto load balancing across additional servers

HedgeDrip Founder

2023 Jan - 2025 Jan

- **Founding & Product Ownership:** Built and launched a digital financial planning platform end-to-end, owning product, engineering, and analytics
- **Product Impact:** Conducted user interviews and behavioral analysis with 20 early adopters, iterating features to improve planning accuracy and usability; 75% reported improved decision confidence

TammyAI Product Engineer

2024 Jan - 2024 Jun

- **User Engagement & Growth:** Increased average session duration from 20 seconds to 2 minutes (+500%), directly contributing to a 10% increase in premium subscriptions
- **Product Collaboration:** Worked closely with the CPO to gather requirements from leadership and end users, translating product vision into scalable solutions and iterating features based on continuous user feedback
- **Video Intelligence Systems:** Built a video Q&A system that generates topic-tagged questions and answers from YouTube content, reducing end-to-end latency from 30 seconds to 3 seconds
- **LLM Feature Optimization:** Engineered celebrity chatbot commentary features serving ~4,000 DAUs, reducing generation time from 4 minutes to 5 seconds through prompt and system-level optimizations
- **Generative Media Features:** Developed custom image generation capabilities for celebrity chatbots, increasing feature-level user retention by 25%
- **User Library & Discovery:** Built a personalized library page for saved and watched videos, supporting approximately 4,000 daily active users

JPMorgan Chase & Co SRE

2023 Jan - 2023 July

- Built predictive models using PyTorch to forecast outages in JPMC's VM unified access gateway to reduce outages
- Modernized legacy infrastructure with Kafka + HDFS-based pipelines, reducing data latency

PROJECTS

Equity research reports

- Conducted equity research and authored an institutional-style report under the guidance of JPMorgan Chase mentors ([link](#))

Machine learning projects

- Designed and deployed several end-to-end machine learning solutions, including a natural language processing chatbot, content-based wine recommendation engine, and collaborative filtering movie recommendation system

SKILLS

Software engineering: Python, Java, JavaScript, R

Frameworks React, Next.js, Spring Boot, Flask

Cloud & Devops AWS, GCP, Docker, Terraform, MLflow

Data Science: Pandas, NumPy, Scikit-learn, PyTorch, NLP, XGBoost, TensorFlow

Finance: Equity research, Options strategies, Portfolio theory